standby mode—I can't see anything but I can hear sounds pretty well. Sometimes I get a '3D device error' too. Has my graphics card gone kaput?

The AGP card that you have is a very low-end 3D-accelerator. Apart from providing basic 3D acceleration, there isn't much else that this card can do for advanced games such as *Need for Speed* and *Quake III*, which incorporate fog effects, specular and dynamic lightning, and tri-linear filtering. Additionally, new graphical programs also use Open GL and other standards, which require much more power in the graphics processor. Hence, this card will allow you to play only very low-end 3D games. If you'd like to play high-end games, buy a graphics card like the GeForce2 or the GeForce3.

Blank screen when playing 3D games

While playing 3D games, the screen suddenly goes blank while I am playing. Sometimes my PC hangs when I close the game and return to my desktop. Why does this happen?

Check to see if the graphics card is being sufficiently cooled. Make sure that there are no loose wires or a cable obstructing the fan on the graphics processor. Remove any dust/grime from the fan. You could try to reduce the quality and resolution settings in the game/Windows environment (display properties). The problem could also be because of incorrect/outdated drivers. See that you are running the latest drivers.

Another reason could be that the motherboard is not able to supply sufficient/correct voltage to the AGP port. In which case, you will have to replace the motherboard.

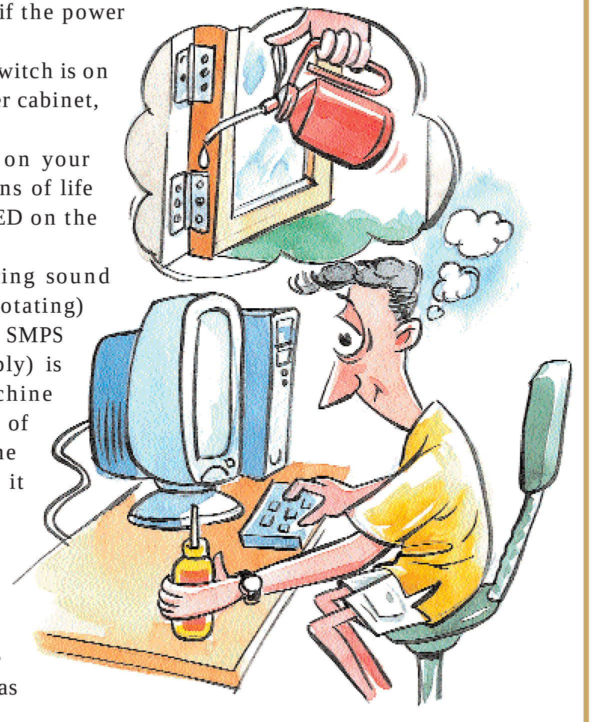
Detonator drivers

What are Detonator drivers and what kind of graphics cards are they meant for?

These drivers are for nVidia graphics processors only. They go hand in hand with DirectX drivers of version 6 and above. These drivers increase the graphics card's performance. Some of the Detonator's features include DVD and HDTV playback, advanced Digital Vibrance Control (DVC) user interface for products that support DVC, and improved FSAA control panels. It also

'My System won't Boot, What do I do?'

No POST (Power On Self Test)



- First and foremost check if the power cable is properly fitted.
- Then check if the toggle switch is on (found behind the computer cabinet, SMPS).
- After this try switching on your machine and check for signs of life (lighting up of the Power LED on the front panel).
- If you can hear a whirring sound (fans and the hard drive rotating) then it means that the SMPS (Switch Mode Power Supply) is fine. If not and the machine sounds dead with no signs of power then try replacing the power cord, otherwise it means that the SMPS is dead.
- If the machine shows signs of power and still does not POST then either some PCI device has come loose or some component has failed.
- You can investigate the problem further if your BIOS emits a peculiar beep code. The following is a common list of beep codes for the popular Award BIOS:

1. One long beep: Memory problem
2. One long, then two short beeps: Video error
3. One long, then three short beeps: Video error
4. Continuous beeping: Memory problem
5. Alternating high-low beeps: CPU not working

POSTs, but hangs too

If1 the system POSTs but hangs on the POST screen, check for the following:

- Bad memory timings in the BIOS or bad RAM. Try changing RAM slots and the RAM module. Revert to default settings in the BIOS.
- Corrupt BIOS (if the POST screen becomes garbled). You will have to change the BIOS EPROM chip.
- Processor is overclocked. Check in the BIOS for the processor and frequency settings. If you are not sure then you can revert to default settings for the BIOS

General guidelines

While it is almost impossible to pin down to a specific problem and a specific solution, there are broad guidelines that you can follow while troubleshooting:

- Boot into Windows using the step-by-step confirmation method if you are using Windows 9x.
- Go through your BIOS settings, and turn everything at the default settings.
- Remove all PCI cards, even the soundcard and try booting only with the video card, hard drive, floppy drive and CD-ROM installed. You then add one card at a time, until the boot problem shows up.
- Long boot times suggest that Windows is most probably trying to resolve an IRQ conflict. You can manually assign IRQs to the PCI slots from the BIOS but do this as a last resort as this can sometimes cause more trouble.
- Try booting into safe mode—if you can, then it's most probably a conflict or a driver-related issue.
- Try uninstalling the last driver you installed or perform a system restore. All versions of Windows after 98 have some form of system restore utility—Windows XP has the best, of course. Try creating as many system restore points when your system is working fine.